

Techy Location-Based Productivity Definition

For this synthetic test, Techy uses a Location Productivity Index (LPI). LPI is an operational proxy based only on location patterns and must not be interpreted as a complete measure of an employee's actual job performance.

Required Association Sequence

For each employee: read the local timestamp and NMEA GGA sentence, classify the GGA latitude/longitude into a named campus location using the Techy Campus NMEA Location Reference, map that location to the productivity weight below, apply the lunch rule, and aggregate all scored intervals.

Scoring Window

Score activity from 09:00 through 17:00 local Techy campus time. The lunch period from 12:00 through 13:00 is excluded from the denominator regardless of location. Each GGA log record represents the 10-minute interval beginning at the local timestamp shown in the log.

Location Weights

Location type	Weight	Rule
Assigned desk	1.00	Focused / individual work
Meeting room	0.90	Collaborative work
Transit / hallway	0.30	Necessary movement between work areas
Campus gym	0.00	Non-work activity during scored working time
Cafeteria	0.00	Break / non-work activity outside excluded lunch
Unknown	0.00	Do not assume productive activity

Calculation

For every scored 10-minute interval, multiply 10 minutes by the weight of the employee's classified location. $LPI = \text{weighted productive minutes} / \text{total scored minutes} \times 100$.

Interpretation

90-100: Very high location alignment

75-89.9: High location alignment

60-74.9: Moderate location alignment

Below 60: Low location alignment

A gym or cafeteria visit during the excluded 12:00-13:00 lunch period does not reduce the score.

A gym or cafeteria visit outside the lunch period receives a weight of 0.00.

When comparing employees, report both the LPI and the major location patterns that explain the difference.